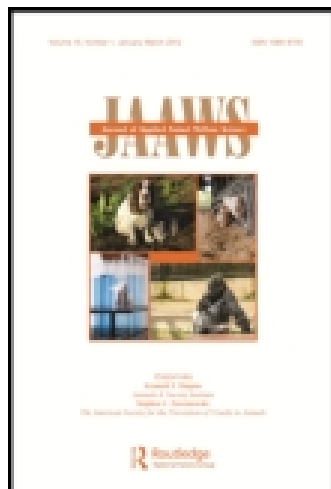




Journal of Applied Animal Welfare Science

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/haaw20>

Morbidity and Mortality of Invertebrates, Amphibians, Reptiles, and Mammals at a Major Exotic Companion Animal Wholesaler

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Published online: 29 May 2014.

To cite this article: Shawn Ashley, Susan Brown, Joel Ledford, Janet Martin, Ann-Elizabeth Nash, Amanda Terry, Tim Tristan & Clifford Warwick (2014) Morbidity and Mortality of Invertebrates, Amphibians, Reptiles, and Mammals at a Major Exotic Companion Animal Wholesaler, Journal of Applied Animal Welfare Science, 17:4, 308-321, DOI: [10.1080/10888705.2014.918511](https://doi.org/10.1080/10888705.2014.918511)

To link to this article: <http://dx.doi.org/10.1080/10888705.2014.918511>

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Morbidity and Mortality of Invertebrates, Amphibians, Reptiles, and Mammals at a Major Exotic Companion Animal Wholesaler

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The authors formally investigated a major international wildlife wholesaler and subsequently confiscated more than 26,400 nonhuman animals of 171 species and types. Approximately 80% of the nonhuman animals were identified as grossly sick, injured, or dead, with the remaining in suspected suboptimal condition. Almost 3,500 deceased or moribund animals (12% of stock), mostly reptiles, were being discarded on a weekly basis. Mortality during the 6-week “stock turnover” period was determined to be 72%. During a 10-day period after confiscation, mortality rates (including euthanasia for humane reasons) for the various taxa were 18% for invertebrates, 44.5% for amphibians, 41.6% for reptiles, and 5.5% for mammals. Causes of morbidity and mortality included cannibalism, crushing, dehydration, emaciation, hypothermic stress, infection, parasite infestation, starvation, overcrowding, stress/injuries, euthanasia on compassionate grounds, and undetermined causes. Contributing factors for disease and injury included poor hygiene; inadequate, unreliable, or inappropriate provision of food, water, heat, and humidity; presumed high levels of stress due to inappropriate housing leading to intraspecific aggression; absent or minimal environmental enrichment; and crowding. Risks for introduction of invasive species through escapes and/or spread of pathogens to naive populations also were identified.

Keywords: *wildlife trade, pet, companion animal, wholesaler, morbidity, mortality*

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During December 2009 and January 2010, a large international exotic companion animal wholesale dealership (U.S. Global Exotics [USGE]) was investigated by veterinarians, biologists, and others under the auspices of Texas state authorities. The entire inventory of nonhuman animals was confiscated and relocated to a controlled facility. The operation followed a 7-month undercover investigation by the People for the Ethical Treatment of Animals (PETA) following concerns regarding animal welfare. A subsequent court hearing revealed that all animals had been treated cruelly, and the Texas Society for the Prevention of Cruelty to Animals and other animal welfare organizations were awarded custody of all the animals. The operation constituted one of the largest and most ambitious exotic animal seizures in U.S. history.

MATERIALS AND METHODS

The primary investigation and management team consisted of 4 veterinarians with exotic animal experience, an animal health technologist with exotic animal experience, a herpetologist, a biologist, and an entomologist. Additional support was provided by a small animal veterinarian, 4 professional zoo curatorial herpetologists, and more than 30 general animal caretakers. Several investigation and general project managers, 10 state police officers, local police officers, and regulatory officials, 4 volunteers for construction and fabrication, and 4 lawyers also were integral to the project.

RESULTS

Morbidity and Mortality Statistics

The investigators found more than 26,400 animals at the facility composed of 171 species or types of invertebrates (30), amphibians (39), reptiles (78), and mammals (24). Approximately 80% of these animals were sick, injured, or deceased. It was uncovered that almost 3,500 deceased or moribund animals (12% of stock), mostly reptiles, were discarded on a weekly basis ($\sim 1.7\%$ per day). This equates to a 72% mortality rate during the standard stock turnover period of 6 weeks. While wholesalers may normally expect a stock turnover of 6 weeks, retailers have shorter turnover periods of approximately 2 weeks (M. Jessop, personal communication, March 12, 2012; C. Steedman, personal communication, March 12, 2012). During a 10-day period under veterinary and other professional supervision, totals and mortality rates (including those euthanized for humane reasons) were calculated (Table 1).

The supplier's staff recorded losses of approximately 500 animals per day. However, following confiscation, we recorded an average loss of 872 animals per day, although this figure includes animals requiring euthanasia. Morbidities and mortalities at the rescue center were determined to be the result of preexisting and underlying diseases attributable to the poor conditions and care provided at the commercial facility prior to confiscation. The mortality rates in these data are particularly concerning given that they reflect losses that occurred during a relatively short period (10 days) as well as those that occurred while the animals were under the care of specialized exotic animal veterinarians and biologists. Figures 1 to 9 provide examples of conditions identified at the wholesaler. Causes of morbidities and mortalities were not subject to methodological recordkeeping by the wholesaler.

TABLE 1
Mortality Rates During a 10-Day Period

<i>Class</i>	<i>Total Number of Animals</i>	<i>Total Number of Mortalities</i>	<i>Mortality Rate (Percentage)</i>
<i>Invertebrate</i>			
arachnids, chilopods, diplopods, crustaceans	2,274	409	18%
<i>Amphibian</i>			
anurans, caudatans	4,567	2,033	44.5%
<i>Reptile</i>			
testudines	6,635	2,453	36.9%
lacertilians	7,755	3,757	48.4%
serpents	1,681	481	28.6%
Cumulative	16,071	6,691	41.6%
<i>Mammal</i>			
rodents, carnivorans, eulipotyphlans, didelphimorphs, diprotodonts, pilosans, primates	3,493	191	5.5%



FIGURE 1 Two emaciated red-tail boas in sparse and understimulating conditions. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)



FIGURE 2 Grossly overcrowded container of baby freshwater turtles. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

Presumptive Causes of Morbidity and Mortality

Presumptive causes of morbidity and mortality (see Table 2) were determined as part of the investigation and management of the animals, through an individual's assessment based on their experience or by group consensus.

Public Health Implications

Given the diversity of species and numbers of animals involved in the investigation, it was anticipated that they could harbor potential animal and zoonotic pathogens. Limited microbiological sampling and parasite detection was conducted on the feces of nine snakes, with samples tested for salmonella, giardia, cryptosporidium, and other endoparasites. All samples were analyzed by the Texas Veterinary Medical Diagnostic Laboratories, and the results are provided in Table 3. Although our investigation presented an opportunity to conduct a general assessment of the possible presence of zoonotic pathogens, time and budgetary limitations allowed for only minimal establishment that the facility did harbor known zoonotic agents (salmonella).



FIGURE 3 Prairie dogs grossly overcrowded and starving. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

Distribution of Traded Animals

USGE was a supplier of exotic animals to innumerable customers (including retail chains and zoological collections) in at least 22 U.S. states, and at least 87 additional customers in 25 countries. The operation was itself supplied by 16 U.S. states, and 22 countries.

The wholesaler had no methodological monitoring for the presence of pathogens in the general environment or in diseased animals. Similarly, necropsies were not carried out to seek to establish the cause of death in animals and the potential for spread of infectious disease.

DISCUSSION

Although it was high, 72% mortality during 6 weeks is similar to that reported in the industry. For example, during judicial proceedings against the dealer, part of the trader's defense cited expert evidence confirming that those mortalities were in accordance with wholesale companion animal industry standards of 70% (J. M. Smith, 2010). Despite already being arguably unacceptably high, wholesaler and retailer mortalities are further compounded by a very high mortality rate in the domestic environment. For example, a recent study of companion reptiles in the United Kingdom showed that premature mortality was 75% within the 1st year in the home, and 81% if losses at retailers are included (Toland, Warwick, & Arena, 2012). The treatment and associated health states of animals at USGE and similar operations directly impacts animal health and survival further down the supply pipeline.



FIGURE 4 Approximately 200 dead green iguanas from a preshipment consignment of approximately 400. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

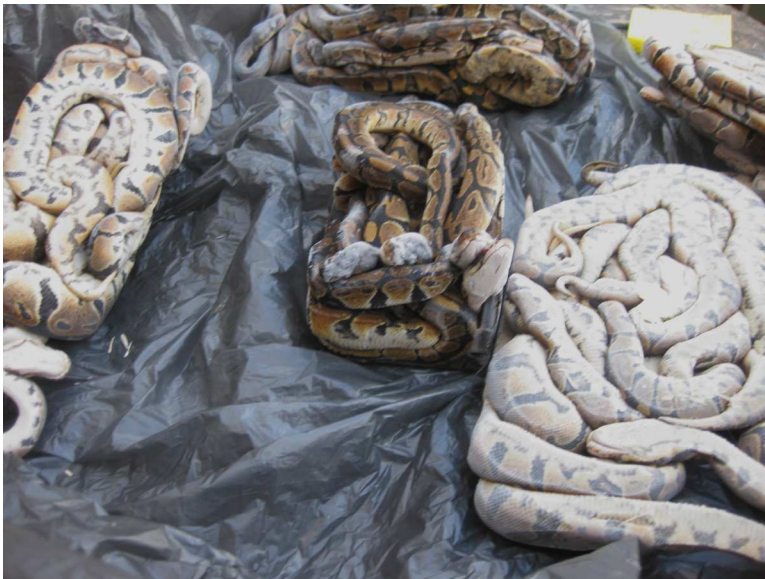


FIGURE 5 Sample selection of deceased animals (royal pythons) removed from wholesaler freezers (common storage method for deceased animals). (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)



FIGURE 6 Frogs crowded into soft-drink bottles. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

The high rates of illness and injury observed in the animals at the facility coincided with their inhumane treatment and lack of adequate care. Contributing factors to this morbidity included poor hygiene; inadequate, unreliable food and water; inappropriate thermal and humidity provisions; likely high stress levels associated with incompatible conspecifics leading to co-occupant aggression; absent or minimal environmental enrichment; overcrowding; and injuries consistent with inhumane handling techniques.

Numerous zoonoses are associated with exotic companion animals (Palmer, Soulsby, Torgerson, & Brown, 2011; Warwick, Arena, Steedman, & Jessop, 2012); for example, approximately 40 zoonoses are linked to reptiles alone (Warwick et al., 2012). Given the generally poor state



FIGURE 7 Sample selection of dead invertebrates (arachnids) and turtles from wholesaler storage enclosures. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

of hygiene at the supply house, we considered that both the animals as well as inanimate surfaces in the commercial facility presented potentially diverse pathogen reservoirs.

In addition to the generally unhygienic conditions and high incidence of disease, no hygiene management or quarantine protocols were in place to control or prevent the spread of disease, either between animals or to humans. Similarly, no systematic quarantine protocols were present for the isolation of newly acquired stock. Accordingly, the facility presented considerable opportunities to disseminate diverse animal species of uncertain health within the United States and globally. It is concerning that so many unhealthy and stressed animals would have experienced extended travel conditions.

Many of the animals facing imminent shipment might also have been carriers of zoonotic diseases to a potentially naïve consumer base. In addition, opportunities were noted for the escape of animals into the local environment, which creates a risk of introducing exotic animal pathogens onto U.S. soil. Further, for nations importing animals from this supplier, many



FIGURE 8 Sample selection of dead (including partly cannibalized) small mammals from wholesaler storage enclosures. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)



FIGURE 9 A diseased sloth. (© People for the Ethical Treatment of Animals [PETA]. Reproduced by permission of PETA. Permission to reuse must be obtained from the rightsholder. Color figure available online.)

TABLE 2
Clinical Signs and Presumptive Causes of Morbidity and Mortality

<i>Class</i>	<i>Type</i>	<i>Signs and Cause(s) of Morbidity and Mortality</i>
Invertebrate	Scorpions, Tarantulas, Millipedes	1, 2, 3, 4
Amphibian	Frogs	1, 2, 3, 4, 5
	Toads	1, 2, 3, 4, 5
	Newts	1, 2, 3, 4, 5
	Salamanders	1, 2, 3, 4, 5
	Caecilians	1, 2, 3, 4, 5
	Sirens	1, 2, 4, 5
Reptile	Freshwater Turtles	1, 2, 3, 4, 5, 6
	Terrestrial Turtles, Tortoises	1, 3, 4
	Lizards	1, 2, 3, 4, 5
	Snakes	1, 2, 3, 4, 5
Small mammal	Hedgehogs	1, 2, 3, 4
	Hamsters	1
	Mice	1, 6
	Rats, Prairie Dogs, Spotted Squirrels, Guinea Pigs	7
	Short-Tailed Opossums	1
	Flying Squirrels	3, 4
	Chinchillas	1, 4, 7
	Wallabies, Sloths, Kinkajous, Coatimundis, Lemurs, Agoutis	1, 2, 3, 5

1. Trauma—environmental overcrowding/injuries, crushing, intraspecific, interspecific, cannibalism
2. Infectious—infection, parasitism
3. Dehydration—nutritional, infection, parasitism, starvation
4. Emaciation—nutritional, infection, parasitism, starvation
5. Environmental—hypothermic stress
6. Euthanasia on compassionate grounds
7. Indeterminate/idiopathic

species harbored pathogens capable of introducing novel diseases into indigenous wildlife and human populations.

However, it is the authors' understanding that these apparently obvious deficiencies reflect normal practices in the exotic animal trade. In the United Kingdom, for example, a "six-point plan" for companion animal sales has been developed to try to address some common disease prevention and control laxities among animal traders. In summary, in the plan, mandates are proposed to ensure that all sick animals receive veterinary treatment; postmortem examinations are conducted within 24 hr when indicated by a veterinarian; regular public health checks are used to survey for the presence of zoonotic agents; and records are maintained for health, morbidity, and mortality; animals supplied, held, and sold; and transactions, including purchaser identification to allow for contact tracing in the event of an outbreak of infectious disease (Animal Protection Agency, 2010).

Quarantine protocols appear highly inconsistent. For example, in many areas of the world, it is a requirement that mammals undergo formal quarantine periods of 30 days to 6 months, notably, but not exclusively, to safeguard against rabies transmission (Centers for Disease

TABLE 3
Laboratory Test Results for Example Animals

Animal	Culture	Specimen	Result
Snake A: <i>Xenochrophis piscator</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake B: <i>Python regius</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake C: <i>Rhadinophis frenatum</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake D: <i>Boa constrictor</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake E: <i>Epicrates cenchria</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake F: <i>Boa constrictor</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake G: <i>Dinodon rufozonatum</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake H: <i>Pantherophis guttatus</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake I: <i>Epicrates cenchria</i>	Salmonella	Feces	<i>Salmonella</i> sp.
Snake A: <i>Xenochrophis piscator</i>	Fecal flotation	Feces	<i>Kalicephalus</i> 3+
Snake B: <i>Python regius</i>	Fecal flotation	Feces	<i>Capillaria</i> sp. 3+
		Feces	<i>Kalicephalus</i> 1+
		Feces	Pinworm 1+
Snake C: <i>Rhadinophis frenatum</i>	Fecal flotation	Feces	<i>Strongyloides</i> ova 3+
		Feces	<i>Capillaria</i> sp. 1+
		Feces	<i>Kalicephalus</i> 1+
Snake D: <i>Boa constrictor</i>	Fecal flotation	Feces	<i>Strongyle</i> ova 3+
		Feces	Pinworm 3+
		Feces	Nematode ova 3+
Snake E: <i>Epicrates cenchria</i>	Fecal flotation	Feces	Pinworm 2+
Snake F: <i>Boa constrictor</i>	Fecal flotation	Feces	Ascarid ova 1+
Snake G: <i>Dinodon rufozonatum</i>	Fecal flotation	Feces	Nematode ova 3+
Snake H: <i>Pantherophis guttatus</i>	Fecal flotation	Feces	Pinworm 1+
Snake I: <i>Epicrates cenchria</i>	Fecal flotation	Feces	Pinworm 3+

Note. Tests requested for samples were for salmonella culture, giardia, cryptosporidium, and fecal flotation. Analysis: Texas Veterinary Medical Diagnostic Laboratories.

Control and Prevention, 2013). However, amphibians and reptiles, despite potentially harboring agricultural animal and zoonotic pathogens, are not routinely quarantined.

Almost 75% of emerging infectious diseases in humans have zoonotic origins (Karesh, Cook, Bennett, & Newcomb, 2005), and there are at least 70 companion animal-related zoonotic pathogens (Warwick et al., 2012). While some zoonotic agents—for example, *Salmonella* spp.—are well known to be associated with amphibians and reptiles (Riley & Chomel, 2005; Warwick et al., 2012), many others are not well documented in the literature. At least 40 companion animal-related zoonoses are associated with amphibians and reptiles.

The viral zoonotic disease monkeypox, causal agent *Poxviridae* sp., was introduced to the United States via imported companion mammals, notably prairie dogs (*Cynomys* sp.), and resulted in a raft of human cases (Centers for Disease Control and Prevention, 2003). Imported African companion reptiles, often tortoises, and migratory birds have been associated with the tickborne pathogen *Ehrlichia ruminantium*, which is responsible for heartwater disease in cattle, sheep, goats, antelope, and buffalo (Center for Food Safety and Public Health/Institute for International Animal Biologics, 2007; Emerging Pathogens Institute, 2013). A very limited viral survey of confiscated imports of mammalian wildlife products revealed important particles including herpes viruses and simian foamy virus (K. M. Smith et al., 2012).

Quarantine guidance to address multispecies scenarios appears elusive. Accordingly, the authors apply here an experiential and consensus approach. Although highly variable according to species and contagion risk factors, quarantine periods for newly acquired animals should generally be recommended at a minimum of 7 days for familiar suppliers or 30 days for unfamiliar suppliers, or longer depending on species and risk.

In the authors' view, good practice has shown that pragmatic quarantine periods for infected animals should be no less than 3 weeks after signs of infection for endotherms (birds and mammals) and no less than 3 months for ectotherms (fishes, amphibians, and reptiles). Possible pathogen latency remains a constant potential threat, making open-ended quarantine an important consideration in various situations.

The exotic companion animal trade appears to be increasingly cited as an important source of human and nonhuman animal infections, with some authors describing the industry as a Trojan horse (Broughton, Verger, & Goffin, 2010; Warwick, 2006). Also, via the exotic companion animal trade, numerous countries may be exposed to atypical pathogenic agents due to questionable border controls. For example, a recent study by the U.S. Government Accountability Office concluded that deficient collaboration was a barrier to effective border control and that agencies should develop a strategy to address the potential risk from imported animals (U.S. Government Accountability Office, 2010).

In addition to its status as a known and potential harbinger of diverse zoonotic-, agricultural-, and wildlife-related pathogens, the global exotic companion animal trade is also of concern because it is a major medium for the introduction of invasive alien species (IAS). Among various species, notable U.S. examples are the Floridian-invading Burmese python (*Python molurus*), which has become a major predator of indigenous wildlife (Harvey et al., 2008), and the Gambian giant pouch rat (*Cricetomys gambianus*), which became an agricultural pest (Wilmer & Hall, 2011). Both of these species emerged as exotic companion animal releases. A recent study in London identified 51 non-native species, subspecies, intergrades, or hybrids of amphibians and reptiles as living wild, all of which are associated with the exotic companion animal trade (Langton, Atkins, & Herbert, 2011).

Exotic companion animals comprise an as-yet imprecise but important IAS component, with (probably highly incomplete) estimates for all invasive species issues indicating hundreds of billions of dollars were lost due to economic damage and the need for management around the world (Strayer, Eviner, Jeschke, & Pace, 2006). Indeed, IAS are regarded to be one of the major threats to European natural biodiversity (Shine et al., 2010).

CONCLUSION

Exotic companion animal wholesale dealerships accumulate large numbers of animals of diverse species from indeterminate sources, each of which presents unique challenges with respect to animal welfare, environmental risk, and potential for pathogen harborage.

This raises serious questions regarding the range of knowledge, experience, and skills required to manage such collections in a manner that protects both the welfare of the animals and their human caretakers. This assessment excludes a secondary extension of these risks to the animals maintained in the same facility or the indigenous fauna, should an escape occur. The core workforce at USGE was typically composed of four casual staff of nonspecialist

backgrounds. The lack of commitment to provide for the basic needs of the animals housed at USGE and to properly train caretaking staff promoted animal neglect and cruelty. Combined with the stress of transportation, handling, and confinement, the situation presented a substantial risk for the propagation of injury and disease.

This case study further demonstrates that although high morbidity and mortality rates may be deemed normal and acceptable by the companion animal industry, such losses are considered unacceptable by state and local authorities.

Greater monitoring of exotic companion animal wholesale dealerships is essential to ensure that appropriate attention is paid to animal welfare and the associated public health and environmental issues in the future. Substantial limitations may be necessary with regard to which animals can be acquired and sold, as well as the numbers of animals and the conditions under which they are housed and maintained.

Animal care standards need to be developed for exotic companion animal dealerships that address husbandry, health care, environmental enrichment, and nutrition. Relevant guidelines exist for animals in captive situations such as accredited zoos and sanctuaries and even livestock management that may act as models to raise and standardize animal welfare and hygiene issues associated with exotic companion animal sellers.

ACKNOWLEDGMENTS

The authors would like to thank the reviewers and Mike Jessop for their helpful and insightful suggestions.

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